

Authoritative Full-Dataset Report

Verbatim contents of `reports/generated_full_dataset/association_summary.txt`, the pipeline-generated snapshot that backs every full-dataset number in the README, deck brief and Q&A pack. Long lines are soft-wrapped for the page; nothing else is altered.

UNSW Marketing Analytics Hackathon 2026 – Reproducible Analysis Report

Audit trail

Source file: `../input/datasets/tuannm3812/kiva-loans-hackathon-data/Kiva_Loans.pkl`
 Rows loaded: 1453846
 Rows with a valid completed outcome: 1453840
 Rows excluded: 6 (`{'negative_duration': 6}`)
 Status among valid rows: `{'funded': 1452203, 'refunded': 1637}` – 'refunded' loans completed funding and are included on the same footing as 'funded' ones; this project models funding speed, not repayment outcome.
 fundraisingDate range: 2016-01-01T02:30:02+00:00 to 2025-12-31T20:30:25+00:00
 Chronological holdout boundary: 2024-01-01
 Period counts: `{'pre_pandemic': 589823, 'post_pandemic': 565474, 'pandemic_disruption': 298549}`
 Funded-within-24h share (valid outcomes): 0.36566265888956145
 Software versions: python=3.12.13, pandas=2.3.3, numpy=2.0.2, scikit_learn=1.6.1, statsmodels=0.14.6

24-Hour Funding Classifier (leakage-safe chronological holdout)

Train rows: 1174953 | Holdout rows: 278887
 Holdout ROC AUC: 0.8997
 Holdout average precision (AP): 0.8301
 Holdout Brier score: 0.1156
 Holdout accuracy: 0.8403
 Holdout class balance: `{'funded_within_24h': 87466, 'not_within_24h': 191421}`

Explanatory Association Summary (robust HC3 standard errors)

Duration model: $\log(1 + \text{funding speed in days}) \sim \text{pre-specified predictors}$, $n = 1453840$ loans with a valid completed outcome. OLS with HC3 heteroskedasticity-robust standard errors.
 Coefficients below describe how each pre-specified predictor is associated with funding speed, holding the other modeled predictors fixed. They describe statistical association, not a causal claim, and every pre-specified predictor is reported regardless of statistical significance.
 Dropped for lacking variation in this sample: `sentiment_available`

- `C(gender_classification)[T.male]`: coefficient 0.4301 (95% CI [0.4253, 0.4349], $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- `C(repaymentInterval)[T.irregularly]`: coefficient -0.3779 (95% CI [-0.3887, -0.3671], $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- `C(repaymentInterval)[T.monthly]`: coefficient -0.1166 (95% CI [-0.1241, -0.1092], $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.

- C(sector)[T.Arts]: coefficient -0.5458 (95% CI $[-0.5724, -0.5193]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- C(sector)[T.Clean Energy]: coefficient -1.0758 (95% CI $[-1.0904, -1.0612]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- C(sector)[T.Clothing]: coefficient 0.2245 (95% CI $[0.2063, 0.2426]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- C(sector)[T.Construction]: coefficient -0.3316 (95% CI $[-0.3639, -0.2992]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- C(sector)[T.Education]: coefficient -1.3785 (95% CI $[-1.4048, -1.3522]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- C(sector)[T.Entertainment]: coefficient -0.6812 (95% CI $[-0.7773, -0.5852]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- C(sector)[T.Food]: coefficient 0.0932 (95% CI $[0.0843, 0.1020]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- C(sector)[T.Health]: coefficient -0.4839 (95% CI $[-0.5112, -0.4565]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- C(sector)[T.Housing]: coefficient -0.1234 (95% CI $[-0.1406, -0.1062]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- C(sector)[T.Manufacturing]: coefficient -0.7615 (95% CI $[-0.7912, -0.7317]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- C(sector)[T.Personal Use]: coefficient 0.1971 (95% CI $[0.1774, 0.2167]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- C(sector)[T.Retail]: coefficient 0.2038 (95% CI $[0.1944, 0.2131]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- C(sector)[T.Reuse & Recycle]: coefficient -0.0575 (95% CI $[-0.0821, -0.0329]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- C(sector)[T.Sanitation & Hygiene]: coefficient -0.3487 (95% CI $[-0.3893, -0.3080]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- C(sector)[T.Services]: coefficient -0.0330 (95% CI $[-0.0482, -0.0177]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- C(sector)[T.Transportation]: coefficient -0.0275 (95% CI $[-0.0519, -0.0032]$, $p=0.0266$) is associated with funding speed, holding the other modeled predictors fixed.
- C(sector)[T.Water]: coefficient -1.6810 (95% CI $[-1.7771, -1.5849]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- C(sector)[T.Wholesale]: coefficient -0.2338 (95% CI $[-0.3279, -0.1398]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- C(region_group)[T.Asia]: coefficient -0.1989 (95% CI $[-0.2060, -0.1917]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- C(region_group)[T.Central America]: coefficient -0.0384 (95% CI $[-0.0531, -0.0238]$, $p=0.0000$) is associated with funding speed, holding the other

- modeled predictors fixed.
- C(region_group)[T.Middle East]: coefficient -1.1881 (95% CI [-1.2329, -1.1434], p=0.0000) is associated with funding speed, holding the other modeled predictors fixed.
 - C(region_group)[T.North America]: coefficient -0.5626 (95% CI [-0.5883, -0.5369], p=0.0000) is associated with funding speed, holding the other modeled predictors fixed.
 - C(region_group)[T.Oceania]: coefficient -0.3081 (95% CI [-0.3312, -0.2850], p=0.0000) is associated with funding speed, holding the other modeled predictors fixed.
 - C(analysis_period)[T.pandemic_disruption]: coefficient 0.5453 (95% CI [0.5366, 0.5539], p=0.0000) is associated with funding speed, holding the other modeled predictors fixed.
 - C(analysis_period)[T.post_pandemic]: coefficient 0.3460 (95% CI [0.3395, 0.3526], p=0.0000) is associated with funding speed, holding the other modeled predictors fixed.
 - C(loan_size_band)[T.medium]: coefficient 0.6660 (95% CI [0.6578, 0.6742], p=0.0000) is associated with funding speed, holding the other modeled predictors fixed.
 - C(loan_size_band)[T.large]: coefficient 0.5924 (95% CI [0.5784, 0.6063], p=0.0000) is associated with funding speed, holding the other modeled predictors fixed.
 - log_loan_amount: coefficient 0.4345 (95% CI [0.4291, 0.4399], p=0.0000) is associated with funding speed, holding the other modeled predictors fixed.
 - lenderRepaymentTerm: coefficient 0.0685 (95% CI [0.0678, 0.0692], p=0.0000) is associated with funding speed, holding the other modeled predictors fixed.
 - is_group_loan: coefficient -0.0043 (95% CI [-0.0119, 0.0033], p=0.2628) is associated with funding speed, holding the other modeled predictors fixed.
 - desc_word_count: coefficient -0.0004 (95% CI [-0.0005, -0.0004], p=0.0000) is associated with funding speed, holding the other modeled predictors fixed.
 - family_mentions_per_100_words: coefficient -0.0226 (95% CI [-0.0253, -0.0199], p=0.0000) is associated with funding speed, holding the other modeled predictors fixed.
 - family_mentions_per_100_words:C(analysis_period)[T.pandemic_disruption]: coefficient 0.0218 (95% CI [0.0190, 0.0245], p=0.0000) is associated with funding speed, holding the other modeled predictors fixed.
 - family_mentions_per_100_words:C(analysis_period)[T.post_pandemic]: coefficient 0.0135 (95% CI [0.0114, 0.0156], p=0.0000) is associated with funding speed, holding the other modeled predictors fixed.
 - family_mentions_per_100_words:C(region_group)[T.Asia]: coefficient 0.0484 (95% CI [0.0462, 0.0506], p=0.0000) is associated with funding speed, holding the other modeled predictors fixed.
 - family_mentions_per_100_words:C(region_group)[T.Central America]: coefficient -0.0531 (95% CI [-0.0598, -0.0465], p=0.0000) is associated with funding speed, holding the other modeled predictors fixed.
 - family_mentions_per_100_words:C(region_group)[T.Middle East]: coefficient -0.0733 (95% CI [-0.0855, -0.0610], p=0.0000) is associated with funding speed, holding the other modeled predictors fixed.
 - family_mentions_per_100_words:C(region_group)[T.North America]: coefficient 0.0393 (95% CI [0.0324, 0.0462], p=0.0000) is associated with funding speed, holding the other modeled predictors fixed.
 - family_mentions_per_100_words:C(region_group)[T.Oceania]: coefficient 0.0311 (95% CI [0.0240, 0.0383], p=0.0000) is associated with funding speed, holding the other modeled predictors fixed.
 - family_mentions_per_100_words:C(loan_size_band)[T.medium]: coefficient -0.0238 (95% CI [-0.0259, -0.0216], p=0.0000) is associated with funding speed, holding the other modeled predictors fixed.

- family_mentions_per_100_words:C(loan_size_band)[T.large]: coefficient -0.0139 (95% CI $[-0.0167, -0.0110]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- family_mentions_per_100_words:C(sector_group)[T.Arts]: coefficient 0.0160 (95% CI $[0.0069, 0.0250]$, $p=0.0006$) is associated with funding speed, holding the other modeled predictors fixed.
- family_mentions_per_100_words:C(sector_group)[T.Clean Energy]: coefficient 0.1249 (95% CI $[0.1191, 0.1306]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- family_mentions_per_100_words:C(sector_group)[T.Clothing]: coefficient 0.0141 (95% CI $[0.0081, 0.0201]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- family_mentions_per_100_words:C(sector_group)[T.Construction]: coefficient 0.0344 (95% CI $[0.0237, 0.0451]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- family_mentions_per_100_words:C(sector_group)[T.Education]: coefficient 0.0690 (95% CI $[0.0642, 0.0739]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- family_mentions_per_100_words:C(sector_group)[T.Food]: coefficient 0.0015 (95% CI $[-0.0014, 0.0045]$, $p=0.3088$) is associated with funding speed, holding the other modeled predictors fixed.
- family_mentions_per_100_words:C(sector_group)[T.Health]: coefficient -0.0244 (95% CI $[-0.0320, -0.0169]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- family_mentions_per_100_words:C(sector_group)[T.Housing]: coefficient -0.0115 (95% CI $[-0.0163, -0.0068]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- family_mentions_per_100_words:C(sector_group)[T.Manufacturing]: coefficient 0.0286 (95% CI $[0.0181, 0.0390]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- family_mentions_per_100_words:C(sector_group)[T.Other]: coefficient 0.0630 (95% CI $[0.0362, 0.0899]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- family_mentions_per_100_words:C(sector_group)[T.Personal Use]: coefficient -0.0744 (95% CI $[-0.0809, -0.0678]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- family_mentions_per_100_words:C(sector_group)[T.Retail]: coefficient -0.0007 (95% CI $[-0.0038, 0.0024]$, $p=0.6526$) is associated with funding speed, holding the other modeled predictors fixed.
- family_mentions_per_100_words:C(sector_group)[T.Reuse & Recycle]: coefficient -0.0093 (95% CI $[-0.0176, -0.0010]$, $p=0.0274$) is associated with funding speed, holding the other modeled predictors fixed.
- family_mentions_per_100_words:C(sector_group)[T.Sanitation & Hygiene]: coefficient -0.0492 (95% CI $[-0.0614, -0.0370]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- family_mentions_per_100_words:C(sector_group)[T.Services]: coefficient 0.0338 (95% CI $[0.0284, 0.0393]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- family_mentions_per_100_words:C(sector_group)[T.Transportation]: coefficient 0.0877 (95% CI $[0.0796, 0.0957]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- family_mentions_per_100_words:C(sector_group)[T.Water]: coefficient 0.3299 (95% CI $[0.2898, 0.3699]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- agency_mentions_per_100_words: coefficient 0.0010 (95% CI $[-0.0009, 0.0028]$, $p=0.3090$) is associated with funding speed, holding the other modeled predictors fixed.

- urgency_mentions_per_100_words: coefficient -0.0627 (95% CI $[-0.0797, -0.0458]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.
- desc_sentiment_compound: coefficient 0.1360 (95% CI $[0.1292, 0.1429]$, $p=0.0000$) is associated with funding speed, holding the other modeled predictors fixed.

24-hour funding model: funded within 24 hours ~ pre-specified predictors, $n = 1453840$ loans with a known 24-hour funding outcome. Binomial GLM (log-odds scale) with HC3 heteroskedasticity-robust standard errors.

Coefficients below describe how each pre-specified predictor is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed. They describe statistical association, not a causal claim, and every pre-specified predictor is reported regardless of statistical significance.

Dropped for lacking variation in this sample: sentiment_available

- C(gender_classification)[T.male]: coefficient -0.9702 (95% CI $[-0.9849, -0.9556]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
- C(repaymentInterval)[T.irregularly]: coefficient 0.7378 (95% CI $[0.7050, 0.7707]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
- C(repaymentInterval)[T.monthly]: coefficient 0.1927 (95% CI $[0.1708, 0.2145]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
- C(sector)[T.Arts]: coefficient 0.8477 (95% CI $[0.7703, 0.9252]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
- C(sector)[T.Clean Energy]: coefficient 3.3372 (95% CI $[3.2753, 3.3991]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
- C(sector)[T.Clothing]: coefficient -0.6034 (95% CI $[-0.6539, -0.5528]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
- C(sector)[T.Construction]: coefficient 0.4811 (95% CI $[0.3815, 0.5807]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
- C(sector)[T.Education]: coefficient 2.8816 (95% CI $[2.8060, 2.9573]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
- C(sector)[T.Entertainment]: coefficient 0.9674 (95% CI $[0.6506, 1.2843]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
- C(sector)[T.Food]: coefficient -0.3432 (95% CI $[-0.3663, -0.3202]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
- C(sector)[T.Health]: coefficient 0.7176 (95% CI $[0.6282, 0.8070]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
- C(sector)[T.Housing]: coefficient 0.3028 (95% CI $[0.2521, 0.3535]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
- C(sector)[T.Manufacturing]: coefficient 1.0449 (95% CI $[0.9440, 1.1458]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
- C(sector)[T.Personal Use]: coefficient -0.9967 (95% CI $[-1.0622, -0.9313]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours,

- holding the other modeled predictors fixed.
- C(sector)[T.Retail]: coefficient -0.5244 (95% CI $[-0.5484, -0.5005]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - C(sector)[T.Reuse & Recycle]: coefficient -0.1879 (95% CI $[-0.2557, -0.1202]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - C(sector)[T.Sanitation & Hygiene]: coefficient 0.5221 (95% CI $[0.3519, 0.6924]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - C(sector)[T.Services]: coefficient -0.1763 (95% CI $[-0.2185, -0.1341]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - C(sector)[T.Transportation]: coefficient -0.1056 (95% CI $[-0.1730, -0.0383]$, $p=0.0021$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - C(sector)[T.Water]: coefficient 2.2668 (95% CI $[1.8897, 2.6439]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - C(sector)[T.Wholesale]: coefficient -0.0030 (95% CI $[-0.3143, 0.3083]$, $p=0.9850$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - C(region_group)[T.Asia]: coefficient 0.3005 (95% CI $[0.2806, 0.3204]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - C(region_group)[T.Central America]: coefficient 0.2119 (95% CI $[0.1652, 0.2585]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - C(region_group)[T.Middle East]: coefficient 2.3605 (95% CI $[2.1620, 2.5590]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - C(region_group)[T.North America]: coefficient 0.4949 (95% CI $[0.4122, 0.5775]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - C(region_group)[T.Oceania]: coefficient 1.1617 (95% CI $[1.0984, 1.2249]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - C(analysis_period)[T.pandemic_disruption]: coefficient -1.1979 (95% CI $[-1.2219, -1.1739]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - C(analysis_period)[T.post_pandemic]: coefficient -0.9147 (95% CI $[-0.9324, -0.8970]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - C(loan_size_band)[T.medium]: coefficient -0.4892 (95% CI $[-0.5109, -0.4674]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - C(loan_size_band)[T.large]: coefficient 0.2996 (95% CI $[0.2578, 0.3413]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - log_loan_amount: coefficient -1.5581 (95% CI $[-1.5750, -1.5413]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - lenderRepaymentTerm: coefficient -0.2111 (95% CI $[-0.2134, -0.2088]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - is_group_loan: coefficient -0.1527 (95% CI $[-0.1749, -0.1306]$, $p=0.0000$) is associated with the log-odds of funding within 24 hours, holding the other

- modeled predictors fixed.
- desc_word_count: coefficient 0.0011 (95% CI [0.0009, 0.0012], p=0.0000) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - family_mentions_per_100_words: coefficient 0.0535 (95% CI [0.0459, 0.0612], p=0.0000) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - family_mentions_per_100_words:C(analysis_period)[T.pandemic_disruption]: coefficient -0.0416 (95% CI [-0.0491, -0.0341], p=0.0000) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - family_mentions_per_100_words:C(analysis_period)[T.post_pandemic]: coefficient -0.0032 (95% CI [-0.0090, 0.0025], p=0.2729) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - family_mentions_per_100_words:C(region_group)[T.Asia]: coefficient -0.0766 (95% CI [-0.0826, -0.0707], p=0.0000) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - family_mentions_per_100_words:C(region_group)[T.Central America]: coefficient 0.0761 (95% CI [0.0559, 0.0962], p=0.0000) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - family_mentions_per_100_words:C(region_group)[T.Middle East]: coefficient 0.1750 (95% CI [0.1238, 0.2263], p=0.0000) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - family_mentions_per_100_words:C(region_group)[T.North America]: coefficient -0.0258 (95% CI [-0.0484, -0.0033], p=0.0247) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - family_mentions_per_100_words:C(region_group)[T.Oceania]: coefficient -0.1101 (95% CI [-0.1294, -0.0908], p=0.0000) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - family_mentions_per_100_words:C(loan_size_band)[T.medium]: coefficient 0.0110 (95% CI [0.0052, 0.0168], p=0.0002) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - family_mentions_per_100_words:C(loan_size_band)[T.large]: coefficient 0.0044 (95% CI [-0.0056, 0.0144], p=0.3882) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - family_mentions_per_100_words:C(sector_group)[T.Arts]: coefficient -0.0516 (95% CI [-0.0786, -0.0245], p=0.0002) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - family_mentions_per_100_words:C(sector_group)[T.Clean Energy]: coefficient -0.2260 (95% CI [-0.2505, -0.2014], p=0.0000) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - family_mentions_per_100_words:C(sector_group)[T.Clothing]: coefficient -0.0444 (95% CI [-0.0612, -0.0276], p=0.0000) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - family_mentions_per_100_words:C(sector_group)[T.Construction]: coefficient -0.1123 (95% CI [-0.1458, -0.0788], p=0.0000) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - family_mentions_per_100_words:C(sector_group)[T.Education]: coefficient -0.1112 (95% CI [-0.1246, -0.0977], p=0.0000) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
 - family_mentions_per_100_words:C(sector_group)[T.Food]: coefficient -0.0135 (95% CI [-0.0211, -0.0059], p=0.0005) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.

- family_mentions_per_100_words:C(sector_group)[T.Health]: coefficient 0.0466 (95% CI [0.0216, 0.0716], p=0.0003) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
- family_mentions_per_100_words:C(sector_group)[T.Housing]: coefficient -0.0355 (95% CI [-0.0495, -0.0216], p=0.0000) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
- family_mentions_per_100_words:C(sector_group)[T.Manufacturing]: coefficient -0.0811 (95% CI [-0.1150, -0.0472], p=0.0000) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
- family_mentions_per_100_words:C(sector_group)[T.Other]: coefficient -0.1503 (95% CI [-0.2401, -0.0605], p=0.0010) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
- family_mentions_per_100_words:C(sector_group)[T.Personal Use]: coefficient 0.2994 (95% CI [0.2784, 0.3204], p=0.0000) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
- family_mentions_per_100_words:C(sector_group)[T.Retail]: coefficient -0.0083 (95% CI [-0.0162, -0.0005], p=0.0375) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
- family_mentions_per_100_words:C(sector_group)[T.Reuse & Recycle]: coefficient -0.0087 (95% CI [-0.0315, 0.0141], p=0.4529) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
- family_mentions_per_100_words:C(sector_group)[T.Sanitation & Hygiene]: coefficient 0.1245 (95% CI [0.0710, 0.1781], p=0.0000) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
- family_mentions_per_100_words:C(sector_group)[T.Services]: coefficient -0.0604 (95% CI [-0.0756, -0.0452], p=0.0000) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
- family_mentions_per_100_words:C(sector_group)[T.Transportation]: coefficient -0.1444 (95% CI [-0.1653, -0.1235], p=0.0000) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
- family_mentions_per_100_words:C(sector_group)[T.Water]: coefficient -0.5040 (95% CI [-0.6495, -0.3585], p=0.0000) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
- agency_mentions_per_100_words: coefficient 0.0226 (95% CI [0.0178, 0.0274], p=0.0000) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
- urgency_mentions_per_100_words: coefficient 0.2219 (95% CI [0.1706, 0.2732], p=0.0000) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.
- desc_sentiment_compound: coefficient -0.1860 (95% CI [-0.2054, -0.1666], p=0.0000) is associated with the log-odds of funding within 24 hours, holding the other modeled predictors fixed.

Cluster-Robust Sensitivity Check

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HC3 standard errors correct for heteroskedasticity but still assume independent observations. Loans sharing a country_name may share unobserved influences (field-partner writing templates, local conditions) that HC3 cannot see. This refits the identical duration and 24-hour formulas with standard errors clustered by country_name instead, and compares which coefficients clear $p < 0.05$ under each – a same-data specification-robustness check: a coefficient significant under both is robust to relaxing HC3's independence assumption, while one significant only under HC3 is not.

Duration model (log funding speed):

- C(gender_classification)[T.male]: HC3 p=0.0000 (significant); clustered p=0.0000 (significant) [same conclusion]
- C(repaymentInterval)[T.irregularly]: HC3 p=0.0000 (significant); clustered p=0.0061 (significant) [same conclusion]
- C(repaymentInterval)[T.monthly]: HC3 p=0.0000 (significant); clustered p=0.0806 (not significant) [CONCLUSION CHANGES]
- C(sector)[T.Arts]: HC3 p=0.0000 (significant); clustered p=0.0001 (significant) [same conclusion]
- C(sector)[T.Clean Energy]: HC3 p=0.0000 (significant); clustered p=0.0000 (significant) [same conclusion]
- C(sector)[T.Clothing]: HC3 p=0.0000 (significant); clustered p=0.0074 (significant) [same conclusion]
- C(sector)[T.Construction]: HC3 p=0.0000 (significant); clustered p=0.0000 (significant) [same conclusion]
- C(sector)[T.Education]: HC3 p=0.0000 (significant); clustered p=0.0000 (significant) [same conclusion]
- C(sector)[T.Entertainment]: HC3 p=0.0000 (significant); clustered p=0.0000 (significant) [same conclusion]
- C(sector)[T.Food]: HC3 p=0.0000 (significant); clustered p=0.2165 (not significant) [CONCLUSION CHANGES]
- C(sector)[T.Health]: HC3 p=0.0000 (significant); clustered p=0.0020 (significant) [same conclusion]
- C(sector)[T.Housing]: HC3 p=0.0000 (significant); clustered p=0.1961 (not significant) [CONCLUSION CHANGES]
- C(sector)[T.Manufacturing]: HC3 p=0.0000 (significant); clustered p=0.0000 (significant) [same conclusion]
- C(sector)[T.Personal Use]: HC3 p=0.0000 (significant); clustered p=0.2810 (not significant) [CONCLUSION CHANGES]
- C(sector)[T.Retail]: HC3 p=0.0000 (significant); clustered p=0.0010 (significant) [same conclusion]
- C(sector)[T.Reuse & Recycle]: HC3 p=0.0000 (significant); clustered p=0.5301 (not significant) [CONCLUSION CHANGES]
- C(sector)[T.Sanitation & Hygiene]: HC3 p=0.0000 (significant); clustered p=0.0000 (significant) [same conclusion]
- C(sector)[T.Services]: HC3 p=0.0000 (significant); clustered p=0.8025 (not significant) [CONCLUSION CHANGES]
- C(sector)[T.Transportation]: HC3 p=0.0266 (significant); clustered p=0.6820 (not significant) [CONCLUSION CHANGES]
- C(sector)[T.Water]: HC3 p=0.0000 (significant); clustered p=0.0000 (significant) [same conclusion]
- C(sector)[T.Wholesale]: HC3 p=0.0000 (significant); clustered p=0.0593 (not significant) [CONCLUSION CHANGES]
- C(region_group)[T.Asia]: HC3 p=0.0000 (significant); clustered p=0.1430 (not significant) [CONCLUSION CHANGES]
- C(region_group)[T.Central America]: HC3 p=0.0000 (significant); clustered p=0.7344 (not significant) [CONCLUSION CHANGES]
- C(region_group)[T.Middle East]: HC3 p=0.0000 (significant); clustered p=0.0000 (significant) [same conclusion]
- C(region_group)[T.North America]: HC3 p=0.0000 (significant); clustered p=0.0002 (significant) [same conclusion]
- C(region_group)[T.Oceania]: HC3 p=0.0000 (significant); clustered p=0.1777 (not significant) [CONCLUSION CHANGES]
- C(analysis_period)[T.pandemic_disruption]: HC3 p=0.0000 (significant); clustered p=0.0014 (significant) [same conclusion]
- C(analysis_period)[T.post_pandemic]: HC3 p=0.0000 (significant); clustered

- p=0.0031 (significant) [same conclusion]
- C(loan_size_band)[T.medium]: HC3 p=0.0000 (significant); clustered p=0.0000 (significant) [same conclusion]
- C(loan_size_band)[T.large]: HC3 p=0.0000 (significant); clustered p=0.0001 (significant) [same conclusion]
- log_loan_amount: HC3 p=0.0000 (significant); clustered p=0.0000 (significant) [same conclusion]
- lenderRepaymentTerm: HC3 p=0.0000 (significant); clustered p=0.0015 (significant) [same conclusion]
- is_group_loan: HC3 p=0.2628 (not significant); clustered p=0.9753 (not significant) [same conclusion]
- desc_word_count: HC3 p=0.0000 (significant); clustered p=0.6565 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words: HC3 p=0.0000 (significant); clustered p=0.1983 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(analysis_period)[T.pandemic_disruption]: HC3 p=0.0000 (significant); clustered p=0.3040 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(analysis_period)[T.post_pandemic]: HC3 p=0.0000 (significant); clustered p=0.2900 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(region_group)[T.Asia]: HC3 p=0.0000 (significant); clustered p=0.0320 (significant) [same conclusion]
- family_mentions_per_100_words:C(region_group)[T.Central America]: HC3 p=0.0000 (significant); clustered p=0.0040 (significant) [same conclusion]
- family_mentions_per_100_words:C(region_group)[T.Middle East]: HC3 p=0.0000 (significant); clustered p=0.0066 (significant) [same conclusion]
- family_mentions_per_100_words:C(region_group)[T.North America]: HC3 p=0.0000 (significant); clustered p=0.0648 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(region_group)[T.Oceania]: HC3 p=0.0000 (significant); clustered p=0.4440 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(loan_size_band)[T.medium]: HC3 p=0.0000 (significant); clustered p=0.1132 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(loan_size_band)[T.large]: HC3 p=0.0000 (significant); clustered p=0.3799 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(sector_group)[T.Arts]: HC3 p=0.0006 (significant); clustered p=0.4092 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(sector_group)[T.Clean Energy]: HC3 p=0.0000 (significant); clustered p=0.0079 (significant) [same conclusion]
- family_mentions_per_100_words:C(sector_group)[T.Clothing]: HC3 p=0.0000 (significant); clustered p=0.4335 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(sector_group)[T.Construction]: HC3 p=0.0000 (significant); clustered p=0.0413 (significant) [same conclusion]
- family_mentions_per_100_words:C(sector_group)[T.Education]: HC3 p=0.0000 (significant); clustered p=0.0329 (significant) [same conclusion]
- family_mentions_per_100_words:C(sector_group)[T.Food]: HC3 p=0.3088 (not significant); clustered p=0.9223 (not significant) [same conclusion]
- family_mentions_per_100_words:C(sector_group)[T.Health]: HC3 p=0.0000 (significant); clustered p=0.3844 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(sector_group)[T.Housing]: HC3 p=0.0000 (significant); clustered p=0.4458 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(sector_group)[T.Manufacturing]: HC3 p=0.0000 (significant); clustered p=0.3940 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(sector_group)[T.Other]: HC3 p=0.0000 (significant); clustered p=0.0101 (significant) [same conclusion]
- family_mentions_per_100_words:C(sector_group)[T.Personal Use]: HC3 p=0.0000 (significant); clustered p=0.1736 (not significant) [CONCLUSION CHANGES]

- family_mentions_per_100_words:C(sector_group)[T.Retail]: HC3 p=0.6526 (not significant); clustered p=0.9642 (not significant) [same conclusion]
- family_mentions_per_100_words:C(sector_group)[T.Reuse & Recycle]: HC3 p=0.0274 (significant); clustered p=0.7076 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(sector_group)[T.Sanitation & Hygiene]: HC3 p=0.0000 (significant); clustered p=0.2382 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(sector_group)[T.Services]: HC3 p=0.0000 (significant); clustered p=0.1105 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(sector_group)[T.Transportation]: HC3 p=0.0000 (significant); clustered p=0.1655 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(sector_group)[T.Water]: HC3 p=0.0000 (significant); clustered p=0.0000 (significant) [same conclusion]
- agency_mentions_per_100_words: HC3 p=0.3090 (not significant); clustered p=0.9291 (not significant) [same conclusion]
- urgency_mentions_per_100_words: HC3 p=0.0000 (significant); clustered p=0.4943 (not significant) [CONCLUSION CHANGES]
- desc_sentiment_compound: HC3 p=0.0000 (significant); clustered p=0.0095 (significant) [same conclusion]

24-hour funding model (log-odds):

- C(gender_classification)[T.male]: HC3 p=0.0000 (significant); clustered p=0.0000 (significant) [same conclusion]
- C(repaymentInterval)[T.irregularly]: HC3 p=0.0000 (significant); clustered p=0.0128 (significant) [same conclusion]
- C(repaymentInterval)[T.monthly]: HC3 p=0.0000 (significant); clustered p=0.3008 (not significant) [CONCLUSION CHANGES]
- C(sector)[T.Arts]: HC3 p=0.0000 (significant); clustered p=0.0134 (significant) [same conclusion]
- C(sector)[T.Clean Energy]: HC3 p=0.0000 (significant); clustered p=0.0000 (significant) [same conclusion]
- C(sector)[T.Clothing]: HC3 p=0.0000 (significant); clustered p=0.0000 (significant) [same conclusion]
- C(sector)[T.Construction]: HC3 p=0.0000 (significant); clustered p=0.0000 (significant) [same conclusion]
- C(sector)[T.Education]: HC3 p=0.0000 (significant); clustered p=0.0000 (significant) [same conclusion]
- C(sector)[T.Entertainment]: HC3 p=0.0000 (significant); clustered p=0.0000 (significant) [same conclusion]
- C(sector)[T.Food]: HC3 p=0.0000 (significant); clustered p=0.0190 (significant) [same conclusion]
- C(sector)[T.Health]: HC3 p=0.0000 (significant); clustered p=0.0007 (significant) [same conclusion]
- C(sector)[T.Housing]: HC3 p=0.0000 (significant); clustered p=0.1322 (not significant) [CONCLUSION CHANGES]
- C(sector)[T.Manufacturing]: HC3 p=0.0000 (significant); clustered p=0.0000 (significant) [same conclusion]
- C(sector)[T.Personal Use]: HC3 p=0.0000 (significant); clustered p=0.1385 (not significant) [CONCLUSION CHANGES]
- C(sector)[T.Retail]: HC3 p=0.0000 (significant); clustered p=0.0000 (significant) [same conclusion]
- C(sector)[T.Reuse & Recycle]: HC3 p=0.0000 (significant); clustered p=0.1569 (not significant) [CONCLUSION CHANGES]
- C(sector)[T.Sanitation & Hygiene]: HC3 p=0.0000 (significant); clustered p=0.0009 (significant) [same conclusion]
- C(sector)[T.Services]: HC3 p=0.0000 (significant); clustered p=0.4259 (not significant) [CONCLUSION CHANGES]

- C(sector)[T.Transportation]: HC3 p=0.0021 (significant); clustered p=0.2788 (not significant) [CONCLUSION CHANGES]
- C(sector)[T.Water]: HC3 p=0.0000 (significant); clustered p=0.0099 (significant) [same conclusion]
- C(sector)[T.Wholesale]: HC3 p=0.9850 (not significant); clustered p=0.9920 (not significant) [same conclusion]
- C(region_group)[T.Asia]: HC3 p=0.0000 (significant); clustered p=0.2312 (not significant) [CONCLUSION CHANGES]
- C(region_group)[T.Central America]: HC3 p=0.0000 (significant); clustered p=0.2668 (not significant) [CONCLUSION CHANGES]
- C(region_group)[T.Middle East]: HC3 p=0.0000 (significant); clustered p=0.0000 (significant) [same conclusion]
- C(region_group)[T.North America]: HC3 p=0.0000 (significant); clustered p=0.1050 (not significant) [CONCLUSION CHANGES]
- C(region_group)[T.Oceania]: HC3 p=0.0000 (significant); clustered p=0.0318 (significant) [same conclusion]
- C(analysis_period)[T.pandemic_disruption]: HC3 p=0.0000 (significant); clustered p=0.0052 (significant) [same conclusion]
- C(analysis_period)[T.post_pandemic]: HC3 p=0.0000 (significant); clustered p=0.0006 (significant) [same conclusion]
- C(loan_size_band)[T.medium]: HC3 p=0.0000 (significant); clustered p=0.0000 (significant) [same conclusion]
- C(loan_size_band)[T.large]: HC3 p=0.0000 (significant); clustered p=0.4250 (not significant) [CONCLUSION CHANGES]
- log_loan_amount: HC3 p=0.0000 (significant); clustered p=0.0000 (significant) [same conclusion]
- lenderRepaymentTerm: HC3 p=0.0000 (significant); clustered p=0.0031 (significant) [same conclusion]
- is_group_loan: HC3 p=0.0000 (significant); clustered p=0.6218 (not significant) [CONCLUSION CHANGES]
- desc_word_count: HC3 p=0.0000 (significant); clustered p=0.5863 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words: HC3 p=0.0000 (significant); clustered p=0.4179 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(analysis_period)[T.pandemic_disruption]: HC3 p=0.0000 (significant); clustered p=0.3204 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(analysis_period)[T.post_pandemic]: HC3 p=0.2729 (not significant); clustered p=0.9019 (not significant) [same conclusion]
- family_mentions_per_100_words:C(region_group)[T.Asia]: HC3 p=0.0000 (significant); clustered p=0.0960 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(region_group)[T.Central America]: HC3 p=0.0000 (significant); clustered p=0.0428 (significant) [same conclusion]
- family_mentions_per_100_words:C(region_group)[T.Middle East]: HC3 p=0.0000 (significant); clustered p=0.0350 (significant) [same conclusion]
- family_mentions_per_100_words:C(region_group)[T.North America]: HC3 p=0.0247 (significant); clustered p=0.4682 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(region_group)[T.Oceania]: HC3 p=0.0000 (significant); clustered p=0.2440 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(loan_size_band)[T.medium]: HC3 p=0.0002 (significant); clustered p=0.6862 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(loan_size_band)[T.large]: HC3 p=0.3882 (not significant); clustered p=0.9164 (not significant) [same conclusion]
- family_mentions_per_100_words:C(sector_group)[T.Arts]: HC3 p=0.0002 (significant); clustered p=0.2637 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(sector_group)[T.Clean Energy]: HC3 p=0.0000

- (significant); clustered p=0.0818 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(sector_group)[T.Clothing]: HC3 p=0.0000 (significant); clustered p=0.2921 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(sector_group)[T.Construction]: HC3 p=0.0000 (significant); clustered p=0.0007 (significant) [same conclusion]
- family_mentions_per_100_words:C(sector_group)[T.Education]: HC3 p=0.0000 (significant); clustered p=0.1123 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(sector_group)[T.Food]: HC3 p=0.0005 (significant); clustered p=0.7228 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(sector_group)[T.Health]: HC3 p=0.0003 (significant); clustered p=0.3220 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(sector_group)[T.Housing]: HC3 p=0.0000 (significant); clustered p=0.2547 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(sector_group)[T.Manufacturing]: HC3 p=0.0000 (significant); clustered p=0.3500 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(sector_group)[T.Other]: HC3 p=0.0010 (significant); clustered p=0.0682 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(sector_group)[T.Personal Use]: HC3 p=0.0000 (significant); clustered p=0.0973 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(sector_group)[T.Retail]: HC3 p=0.0375 (significant); clustered p=0.8200 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(sector_group)[T.Reuse & Recycle]: HC3 p=0.4529 (not significant); clustered p=0.8421 (not significant) [same conclusion]
- family_mentions_per_100_words:C(sector_group)[T.Sanitation & Hygiene]: HC3 p=0.0000 (significant); clustered p=0.1325 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(sector_group)[T.Services]: HC3 p=0.0000 (significant); clustered p=0.1894 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(sector_group)[T.Transportation]: HC3 p=0.0000 (significant); clustered p=0.1821 (not significant) [CONCLUSION CHANGES]
- family_mentions_per_100_words:C(sector_group)[T.Water]: HC3 p=0.0000 (significant); clustered p=0.0258 (significant) [same conclusion]
- agency_mentions_per_100_words: HC3 p=0.0000 (significant); clustered p=0.2045 (not significant) [CONCLUSION CHANGES]
- urgency_mentions_per_100_words: HC3 p=0.0000 (significant); clustered p=0.2233 (not significant) [CONCLUSION CHANGES]
- desc_sentiment_compound: HC3 p=0.0000 (significant); clustered p=0.0190 (significant) [same conclusion]

Average Within-Region Family-Framing Slopes

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An interaction coefficient only tests whether a region_group level's family_mentions_per_100_words slope differs from the reference level's. It does NOT test whether the focal measure is associated with the outcome WITHIN that group – and because the focal term is interacted with several moderators at once, 'main effect + group term' is the slope only at the other moderators' reference levels, one unrepresentative cell. This section reports the correct quantity: each group's slope averaged over that group's own observed composition of the other moderators, as a weighted linear contrast, with standard errors and 95% confidence intervals under both HC3 and country_name-clustered covariance. The clustered p-value and interval use statsmodels' normal reference, which is only trustworthy with many clusters – and a group's slope is identified only by the clusters inside that group. Each row therefore also reports a HEURISTIC few-cluster sensitivity reference: a t distribution with (clusters in group – 1) degrees of freedom on the unchanged clustered SE. Cameron & Miller (2015, sec. VI) discuss $t(G-1)$ as the minimum

improvement over the normal reference but warn that even it can over-reject with the standard CRVE; their better-calibrated small-cluster procedures are not implemented here, and substituting the clusters inside one group for the fit's full cluster count is a further conservative heuristic, not a derived result. These p-values and intervals are a sensitivity screen, never grounds for claiming significance: failing the screen downgrades a result to descriptive; passing it must NOT be read as statistical support. For a two-cluster group the reference is $t(1)$, whose 95% critical value is 12.7 instead of 1.96. For a single-cluster group, between-cluster uncertainty is not estimable and no few-cluster values are reported at all.

Sign conventions are OPPOSITE between the two models: the duration model is $\log(1 + \text{days})$, so NEGATIVE = faster funding; the 24-hour model is log-odds of funding within 24h, so POSITIVE = faster.

A group's slope is identified only by the clusters within it; a group containing very few clusters cannot separate the focal association from whatever else is idiosyncratic about those specific clusters, and its estimate pools the whole group – it is not evidence about any individual constituent cluster.

Duration model (log funding speed; NEGATIVE = faster):

- Africa (610368 loans, 27 country_name cluster(s)): estimate=-0.0158; HC3 se=0.0008 95% CI [-0.0173, -0.0142] p=0.0000 | clustered se=0.0160 95% CI [-0.0471, 0.0155] p=0.3225 | few-cluster $t(26)$ 95% CI [-0.0486, 0.0170] p=0.3316 [not significant]
- Asia (738191 loans, 12 country_name cluster(s)): estimate=0.0234; HC3 se=0.0007 95% CI [0.0220, 0.0249] p=0.0000 | clustered se=0.0136 95% CI [-0.0032, 0.0501] p=0.0846 | few-cluster $t(11)$ 95% CI [-0.0065, 0.0534] p=0.1125 [not significant]
- Central America (59391 loans, 2 country_name cluster(s)): estimate=-0.0742; HC3 se=0.0033 95% CI [-0.0806, -0.0678] p=0.0000 | clustered se=0.0070 95% CI [-0.0879, -0.0605] p=0.0000 | few-cluster $t(1)$ 95% CI [-0.1629, 0.0145] p=0.0597 [significant under the normal reference only – NOT under the few-cluster reference; descriptive, not statistically supported]
- Middle East (14946 loans, 2 country_name cluster(s)): estimate=-0.0729; HC3 se=0.0061 95% CI [-0.0849, -0.0609] p=0.0000 | clustered se=0.0140 95% CI [-0.1003, -0.0456] p=0.0000 | few-cluster $t(1)$ 95% CI [-0.2502, 0.1043] p=0.1203 [significant under the normal reference only – NOT under the few-cluster reference; descriptive, not statistically supported]
- North America (7559 loans, 1 country_name cluster(s)): estimate=0.0127; HC3 se=0.0034 95% CI [0.0061, 0.0193] p=0.0002 | clustered se=0.0049 95% CI [0.0031, 0.0223] p=0.0094 | few-cluster reference: not estimable (single cluster – between-cluster uncertainty undefined) [significant under HC3 and normal-reference clustering, but identified by a single cluster – descriptive only]
- Oceania (23385 loans, 4 country_name cluster(s)): estimate=0.0050; HC3 se=0.0035 95% CI [-0.0019, 0.0119] p=0.1584 | clustered se=0.0376 95% CI [-0.0688, 0.0787] p=0.8949 | few-cluster $t(3)$ 95% CI [-0.1148, 0.1247] p=0.9032 [not significant]

24-hour funding model (log-odds; POSITIVE = faster):

- Africa (610368 loans, 27 country_name cluster(s)): estimate=0.0271; HC3 se=0.0023 95% CI [0.0225, 0.0316] p=0.0000 | clustered se=0.0327 95% CI [-0.0370, 0.0912] p=0.4075 | few-cluster $t(26)$ 95% CI [-0.0401, 0.0943] p=0.4150 [not significant]
- Asia (738191 loans, 12 country_name cluster(s)): estimate=-0.0304; HC3 se=0.0019 95% CI [-0.0341, -0.0267] p=0.0000 | clustered se=0.0285 95% CI

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[-0.0862, 0.0254] p=0.2860 | few-cluster t(11) 95% CI [-0.0931, 0.0323]  
p=0.3088 [not significant]  
- Central America (59391 loans, 2 country_name cluster(s)): estimate=0.1025; HC3  
se=0.0099 95% CI [0.0831, 0.1220] p=0.0000 | clustered se=0.0237 95% CI  
[0.0561, 0.1490] p=0.0000 | few-cluster t(1) 95% CI [-0.1986, 0.4037]  
p=0.1446 [significant under the normal reference only - NOT under the  
few-cluster reference; descriptive, not statistically supported]  
- Middle East (14946 loans, 2 country_name cluster(s)): estimate=0.1753; HC3  
se=0.0257 95% CI [0.1249, 0.2258] p=0.0000 | clustered se=0.0608 95% CI  
[0.0561, 0.2946] p=0.0040 | few-cluster t(1) 95% CI [-0.5978, 0.9485]  
p=0.2127 [significant under the normal reference only - NOT under the  
few-cluster reference; descriptive, not statistically supported]  
- North America (7559 loans, 1 country_name cluster(s)): estimate=0.0032; HC3  
se=0.0111 95% CI [-0.0186, 0.0250] p=0.7759 | clustered se=0.0071 95% CI  
[-0.0108, 0.0171] p=0.6570 | few-cluster reference: not estimable (single  
cluster - between-cluster uncertainty undefined) [not significant]  
- Oceania (23385 loans, 4 country_name cluster(s)): estimate=-0.0723; HC3  
se=0.0095 95% CI [-0.0908, -0.0537] p=0.0000 | clustered se=0.0912 95% CI  
[-0.2511, 0.1066] p=0.4284 | few-cluster t(3) 95% CI [-0.3626, 0.2181]  
p=0.4862 [not significant]
```